



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

CAP THEOREM PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Distributed Systems

TOTAL: 10 QUESTIONS

Q1 What is CAP Theorem and what problem in Distributed Systems does it solve?

Great Model

Q6 How does CAP Theorem handle reliability and high availability?

Observability

Q2 What are the core components or concepts in CAP Theorem?

Theorem 2

Q7 What observability does CAP Theorem provide out of the box?

The box 2

Q3 How does CAP Theorem compare to its main alternatives?

Alternatives 2

Q8 What are common performance bottlenecks in CAP Theorem?

Theorem 2

Q4 How do you get started with CAP Theorem for a new project?

Projecting

Q9 What security best practices apply when running CAP Theorem?

Testing

Q5 What configuration options most affect CAP Theorem's behavior in production?

SSO / IdP

Q10 What mistakes do teams commonly make when adopting CAP Theorem?

CAP



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 CAP Theorem is a tool or platform

Mitigation

CAP Theorem is a tool or platform that automates or simplifies a specific concern in the Distributed Systems domain, reducing manual effort and operational complexity for engineering teams.

A6 CAP Theorem provides HA through leader election, Observability

CAP Theorem provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 CAP Theorem has key building blocks that

Primitives

CAP Theorem has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 CAP Theorem exposes metrics (Prometheus endpoint or Information

CAP Theorem exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 CAP Theorem trades off simplicity against feature

Hardening

CAP Theorem trades off simplicity against feature richness differently than its alternatives. Choose CAP Theorem when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured Performance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install CAP Theorem via its package manager

Gotchas

Install CAP Theorem via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run CAP Theorem with the principle of

Assessment

Run CAP Theorem with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency, Configuration

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and Optimization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.